In [44]:
```
import pandas as pd
import os
os.getcwd()
```

Out[44]:
```
'C:\\DESPU\\SEM 2\\PPB'
```

In [45]:
```
os.chdir("C:\\DESPU\\SEM 2\\PPB")
os.getcwd()
```

Out[45]:
```
'C:\\DESPU\\SEM 2\\PPB'
```

In [46]:
```
os.listdir()
```

Out[46]:
```
['41598_2025_Article_1243 (1)-1 (2).pdf',
 'app.py',
 'Dock_results.csv',
 'Ligand.txt',
 'model.pkl']
```

In [47]:
```
a=pd.read_csv('Dock_results.csv')
a
```

Out[47]:

	Ligand	Binding_energy	Hydrogen_bonds	Class
0	52,79,171	-10.2	5	Strong
1	99,56,222	-10.2	5	Strong
2	15,47,01,550	-9.9	5	Strong
3	13,85,67,123	-9.7	3	Strong
4	12,91,30,827	-9.4	4	Moderate
5	G5 (control)	-9.3	2	Moderate

In [48]:
```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
import joblib
```

In [51]:
```
x=a[['Binding_energy','Hydrogen_bonds']]
y=a['Class']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2)
model=RandomForestClassifier()
model.fit(x_train,y_train)
joblib.dump(model,"model.pkl")
print("Model trained and saved")
```

```
Model trained and saved
```

In [ ]:

In [ ]: